

Experiment - 06

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 Management System

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[A.]

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

<u>sale_date</u>	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output

<u>sale_date</u>	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20
2020-05-04	-1

Ans.

SELECT

sale_date,

SUM(CASE WHEN fruit = 'apples' THEN sold_num ELSE 0 END)

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SUM(CASE WHEN fruit = 'oranges' THEN sold_num ELSE 0 END) AS diff

FROM Sales

ORDER BY sale_date;

[B.]

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- **Root:** A node that does not have any parent.
- **Leaf:** A node that has a parent but does not have any children.
- **Inner:** A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tree		Expected Output	
id	p_id	id	Type
1	NULL	1	Root
2	1	2	Inner
3	1	3	Leaf
4	2	4	Leaf
5	2	5	Leaf

Ans:

SELECT

id,

CASE

WHEN p_id IS NULL THEN 'Root'

WHEN id IN (SELECT p_id FROM Tree) THEN 'Inner'

ELSE 'Leaf'

END AS Type

FROM Tree

ORDER BY id;